FCAI – Image Processing Project

Project Documentation

Course: CS213 | OOP 2026–2027
Section: [S1,S2]
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1. Project Overview

This project is a console-based image processing application written in C++. It uses the provided Image_Class.h library to load an image, apply a selected filter, and save the processed image to a user-specified filename.

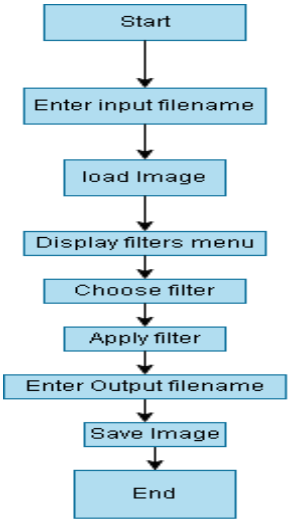
Main workflow:

1. The user enters the input image filename.
2. The program loads the image.
3. The program displays a unified menu of available filters.
4. The user chooses a filter and the program applies it.
5. The user enters an output filename and the program saves the result.

2. Team Members and Responsibilities

Full Name	Student ID	Section	Assigned Filters / Tasks
Mohamed Ibrahim	[20240459]	[S1, S2]	Filters 1 ,5 ,9 ,13
Anas Rafaat	[20250093]	[S1, S2]	Filters 2 ,6 ,10 ,14
Omar Ahmed	[20250420]	[S1, S2]	Filters 3 ,7 ,11 ,15
Eyad Ahmed	[20250897]	[S1, S2]	Filters 4 ,8 ,12 ,14

3. System Diagram



4. System Breakdown

Component	Description
Input	Read the input image filename from the user.
Image Loading	Create/load an Image object using Image_Class.h.
Filter Menu	Display the available filters and read the user's choice.
Filter Processing	Call the function for the selected filter.
Output	Read the output filename and save the modified image.
Validation	Check invalid menu choices, image-loading errors, and output filename/extension where applicable.

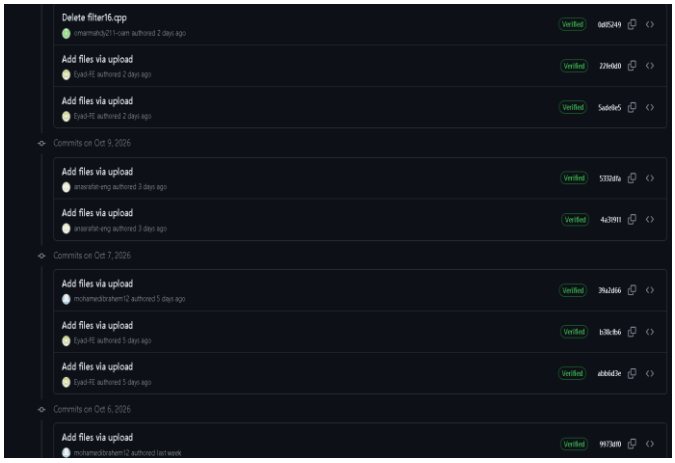
5. Filters Implemented

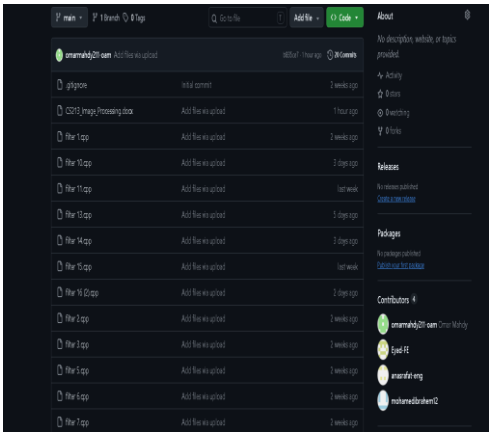
Complete this table with the actual filters included in the final integrated program. Use the exact filter names shown in your menu.

Filter No.	Filter Name	Team Member	Tested?
1	[Gray scale]	[Mohamed Ibrahim]	[Yes]
2	[Black& white]	[Anas Raafat]	[Yes]
3	[Invert Image]	[Omar Ahmed]	[Yes]
4	[Adding a frame]	[Eyad Ahmed]	[Yes]
5	[Flip Image]	[Mohamed Ibrahim]	[Yes]
6	[Rotate Image]	[Anas Raafat]	[Yes]
7	[Darken & lighten]	[Omar Ahmed]	[Yes]
8	[Resizing Image]	[Eyad Ahmed]	[Yes]
9	[Marge Images]	[Mohamed Ibrahim]	[Yes]
10	[Detect Image edges]	[Anas Raafat]	[Yes]
11	[Crop Image]	[Omar Ahmed]	[Yes]
12	[Blur Image]	[Eyad Ahmed]	[Yes]
13	[Natural sunlight]	[Mohamed Ibrahim]	[Yes]
14	[TV Image]	[Anas Raafat]	[Yes]
15	[Purple Effect]	[Omar Ahmed]	[Yes]
16	[Infrared Filter]	[Eyad Ahmed]	[Yes]

6. GitHub Repository Evidence

Repository link: [omarmahdy211-oam/Assignment-OOP-]






```cpp
void filter_11 (Image& image) {
    int x ,y ;
    cout << "Enter the x and y coordinates of the image:";
    cin >> x >> y;

    int new_width ,new_height ;
    cout << "Enter the width and height of the image:";
    cin >> new_width >> new_height;

    Image new_image(new_width,new_height);

    if( x + new_width <= image.width && y + new_height <= image.height){
        for (int i = x ; i <  (x+new_width) ; i++) {
            for (int j = y ; j <  (y+new_height) ; j++) {
                new_image(i-x , j-y ,0) =image(i , j ,0);
                new_image(i-x , j-y ,1) =image(i , j ,1);
                new_image(i-x , j-y ,2) =image(i , j ,2);
            }
        }
    }
    else {
        cout << "Invalid crop dimensions!" << endl;
        return  ;
    }
    image = new_image ;
}
```

```cpp
#include <iostream>
#include "Image_Class.h"
using namespace std;


void filter_14(Image& image) {
    for (int i = 0; i < image.width; ++i) {
        for (int j = 0; j < image.height; ++j) {
            if (i % 3 == 0) {
                image(j, i, 0) = image(j, i, 0) * 0.4;
                image(j, i, 1) = image(j, i, 1) * 0.4;
                image(j, i, 2) = image(j, i, 2) * 0.4;
            } else {
                image(j, i, 0) = image(j, i, 0) * 0.85;
                image(j, i, 2) = image(j, i, 2) * 0.90;
            }
        }
    }
}
```

7. Challenges and Lessons Learned

Best part of the project: The Menu

Hardest part of the project: Some filters such as filters 12 & 10

How the team solved the challenges: We communicated with each other on Discord discussed the problems we faced ,and worked to solve them

8. Demonstration Video

Video URL (https://drive.google.com/file/d/1YlsFHIK3YH3uCqkvvIT-rcp22uUpNUU3/view?usp=drivesdk)

